

Ben Seidel
benseidel
benseidel@vt.edu
ECE 5984 Project 1

I had kit number 35. My kit was able to pass every single test from the F26_DeviceKitTest.ino test. I was able to visually verify that the tests were passed. There were also no problems that I encountered when installing the accelerometer library or when compiling/executing the test program.

My approach to designing the sensor display program was to first go into the links given and find out what the tft LCD header file was able to do as well as google the appropriate calls to get sensor data. I then played around with getting the values and screens to display in a proper way. After researching the appropriate calls to make for buttons and sensor data. I made the screens individually. I tested each screen one by one and would compile and execute after every small change that I made. After I had the code working for the screens, I formatted how the screens would change in a similar way to I did in embedded, where we made an enum with different screen names along with display and updateScreen functions. Some problems that I encountered while completing this project were finding out what different library functions do, reading the accelerometer data (I didn't set up the init variables correctly), and printing the numbers (I had to look through the code given for the device kit test to find the function to display floats/numbers). I had no trouble setting up the Arduino IDE. Overall, all parts of the specification were met.

Some things that I learned from this project were the tft LCD header file functions where you can display different fonts, colors, and strings on the WIO terminal LCD. I also learned about the different button calls and sensor pins that had to be called to read from the WIO terminal. One thing that I would improve on this project is for you to go over the tft library a bit in class briefly before giving us documentation.